

Experiment No: 2

Title: Write a Python Program to Solve the 8-Queen Problem

Aim

To develop a Python program to solve the 8-Queen Problem using the Backtracking algorithm by placing eight queens on a chessboard such that no two queens attack each other.

Objective

To implement the Backtracking algorithm for finding a valid arrangement of eight queens on an 8×8 chessboard while satisfying all the constraints.

Algorithm

- **Step 1:** Define the size of the chessboard as 8×8 .
- **Step 2:** Start placing queens from the first row.
- **Step 3:** Check whether the current position is safe.
- **Step 4:** If the position is safe, place the queen.
- **Step 5:** Move to the next row and repeat the process.
- **Step 6:** If no safe position is found, backtrack to the previous row.
- **Step 7:** Continue until all eight queens are placed.
- **Step 8:** Display the final arrangement of queens.

Program

```
N = 8
```

```
board = [[0 for _ in range(N)] for _ in range(N)]
```

```
def is_safe(row, col):
```

```
    # Check left side
```

```
    for i in range(col):
```

```
        if board[row][i] == 1:
```

```
            return False
```

```
    # Check upper diagonal
```

```
    i, j = row, col
```

```
while i >= 0 and j >= 0:
```

```
    if board[i][j] == 1:
```

```
        return False
```

```
    i -= 1
```

```
    j -= 1
```

```
# Check lower diagonal
```

```
i, j = row, col
```

```
while i < N and j >= 0:
```

```
    if board[i][j] == 1:
```

```
        return False
```

```
    i += 1
```

```
    j -= 1
```

```
return True
```

```
def solve(col):
```

```
    if col >= N:
```

```
        return True
```

```
for row in range(N):
```

```
    if is_safe(row, col):
```

```
        board[row][col] = 1
```

```
        if solve(col + 1):
```

```
            return True
```

```
        board[row][col] = 0
```

```
return False
```

```
if solve(0):
```

```
    print("Solution Found:\n")
```

```
    for row in board:
```

```
        print(*row)
```

```
else:
```

```
    print("No Solution Exists")
```

Sample Input

No user input required.

Sample Output

Solution Found:

```
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0
0 0 0 0 0 1 0 0
0 0 1 0 0 0 0 0
|
```

Result

Thus, the Python program to solve the **8-Queen Problem** using the **Backtracking algorithm** was executed successfully, and a valid arrangement of eight queens was obtained.